COURSE DESCRIPTION FORM: RB-XXXX Embedded Control of Robotics

INSTITUTION FAST School of Computing, National University of Computer and Emerging Sciences, Islamabad

PROGRAM TO BE EVALUATED BS-CS

Course Description

Course Code	CS-4095
Course Title	Embedded Control of Robotics
Credit Hours	3+0
Prerequisites by Course(s) and Topics	
Grading Policy	Absolute grading
Policy about missed assessment items in the course	Retake of missed assessment items (other than sessional/ final exam) will not be held. For a missed sessional / final exam, an exam retake/ pretake application along with necessary evidence are required to be submitted to the department secretary. The examination assessment and retake committee decides the exam retake/ pretake cases.
Course Plagiarism Policy	Plagiarism in project or sessional/ final exam may result in F grade in the course. Plagiarism in an assignment will result in zero marks in the whole assignments category.
Assessment Instruments with Weights (homework, quizzes, midterms, final, programming assignments, lab work, etc.)	***100% Theory*** Assessment Items

Assessment Item	Number	Weight (%)
Assignments	3-5	10
Project	1	10
Sessional 1	1	15
Sessional 2	1	15
Quizzes	4-6	10
Final Exam	1	40

Course Instructors	Dr. Imran Ashraf
Lab Instructors (if any)	
Course Coordinator	Dr. Imran Ashraf
URL (if any)	
Current Catalog Description	This course aims to provide necessary background knowledge and understanding of embedded control techniques to real-world industrial problems within the domain of robotics and beyond. This immersive course is designed to demystify the fascinating world of control

	systems and empower you with the knowledge and skills to navigate this well-established field. Through a blend of theoretical insights, hands-on exercises, and real-world examples, you will not only gain proficiency in fundamentals of control theory but also discover how control systems are commissioned in industry. By the end of this course, you will be able to realize basic embedded controllers.
Textbook	1. Embedded Digital Control with Microcontrollers, 1st Edition, 2021, Cem Unsalan, Duygun E. Barkana, H. Deniz Gurhan, Wiley-IEEE Press
Reference Material	1. Classical and Modern Controls with Microcontrollers, Design, Implementation, and Applications, Springer, 2019, Ying Bai, Zvi S. Roth 2. Process Control in Practice, 1st Edition, 2023, Tore Hägglund, Walter de Gruyter GmbH
Course Learning Outcomes	**A. Course Learning Outcomes (CLOs)** After completion of the course, the students shall be able to: 1. **Describe** basic control system theory and applications. 2. **Analyze** control systems for stability and performance. 3. **Design** controllers for robotic systems for stability and performance enhancement.

PLO 1	Computing Knowledge	Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems
PLO 2	Problem Analysis	Identify, formulate, research literature, and analyze complex computing problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences.
PLO 3	Design/Develop Solutions	Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
PLO 4	Investigation & Experimentation	Conduct investigation of complex computing problems using research-based knowledge and research-based methods.
PLO 5	Modern Tool Usage	Create, select, and apply appropriate techniques, resources and modern computing tools, including prediction and modelling for complex computing problems.
PLO 6	Society Responsibility	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues relevant to context of complex computing problems.
PLO 7	Environment and Sustainability	Apply ethical principles and commit to professional ethics and responsibilities and norms of computing practice.
PLO 8	Ethics	Apply ethical principles and commit to professional

		ethics and responsibilities and norms of computing practice.
PLO 9	Individual and Teamwork	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
PLO 10	Communication	Communicate effectively on complex computing activities with the computing community and with society at large
PLO 11	Project Management and Finance	Demonstrate knowledge and understanding of management principles and economic decision making and apply these to one's own work as a member of a team.
PLO 12	Lifelong Learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological changes.

C. Mapping of CLOs to PLOs
(CLO: Course Learning Outcome, PLOs: Program Learning Outcomes)

		PLOs											
		1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
CLOs	1	✓											
	2		✓										
	3			✓		✓							

Topics covered in the course with number of lectures on each topic (assume 15-week instruction and 3 contact hours per week)

1. Topics to be covered:

List of Topics	No. of Weeks	Contact Hours	CLO(s)
Introduction, System, Control system building blocks, Control system terminology, Embedded Control, Math recap for control system, Python recap for control and embedded system, Intro to micro python	**1**	**3**	**1**
Analogue signals, Fundamentals of Digital Signals, Mathematical definition, representation in code, standard digital signals	**1**	**3**	**1**
Conversion Between Analog and Digital Forms, ADC, Discretization, Quantization, DAC, mathematical representation, code	**1**	**3**	**1**

	implementation			
	Differential Equations, Laplace Transform, Transfer functions, block diagrams, signal flow graphs	**1**	**3**	**1**
	Fundamentals of Digital System control, Difference equations, representatoin of digital systems, properties of digital systems, z-transform and LTI system	**1**	**3**	**1**
	Stability analysis in python and micro-python, Acquiring Digital Signals from the Microcontroller, Processing Offline/Real-Time Data	**1**	**3**	**2**
	Control system performance metrics	**1**	**3**	**2**
	Transfer function based control system analysis, root locus	**1**	**3**	**2**
	Transfer function based control system design, Controller types, PID Controllers, Digital version of PID Controllers, PID in continous time, PID in discrete time	**1**	**3**	**3**
	PID controller tuning methods	**1**	**3**	**3**
	State-space models, differential/difference equations to state space and back, transfer function to state space and back	**1**	**3**	**1**
	State-space based controller analysis (stability and performance analysis)	**1**	**3**	**2**
	State-space based controller design (relocation of eigen values)	**1**	**3**	**3**
	realization of controllers (sampling, quantization)	**1**	**3**	
	Advanced topics: fuzzy control, neural network-based controllers, model predictive control, non-linear control, robust control, distributed control	**1**	**3**	
	Total	**15**	**45**	

Laboratory Projects/ Experiments Done in the Course	NA			
Class Time Spent (in percentage)	**Theory**	**Problem Analysis**	**Solution Design**	**Social and Ethical Issues**
	20	35	40	5
Oral and Written Communications	Every student is required to submit at least 1 written reports/implementation of the semester project and to make oral presentation/demo of typically 10 minute's duration.			